

# Layla Aure

## SUMMARY:

Computer Science graduate with experience developing full-stack applications, developer tools, and AI-integrated software. Built software used by 250+ users, including a Python attendance system and a browser-based coding platform. Strong foundation in software engineering, systems, and scalable application development.

## EDUCATION:

### James Madison University

May 2026

*Bachelor of Science - Computer Science*

GPA: 3.622 | President's List (Spring 2026) | Dean's List (All Semesters)

**Relevant Coursework:** Algorithms & Data Structures, Discrete Structures, Computer Systems, Software Engineering, Database Systems, Applied Algorithms, Programming Languages, Operating Systems, Web Development

## TECHNICAL SKILLS:

**Programming Languages:** Python, Java, C, C++, C#, JavaScript, SQL, Ruby, HTML/CSS

**Web & Frameworks:** React, Vue.js, Node.js, Express.js, Flask, REST APIs, Vite

**Tools & Technologies:** Git, GitHub, GitHub Actions, Linux, Bash, Docker, PostgreSQL, MySQL, AWS (S3, CloudFront), Cloudflare, Nginx, PM2, OIDC/IAM, Google Sheets API, PyInstaller

**Technical Concepts:** Full-Stack Development, API Integration, Client-Server Architecture, CI/CD, Object-Oriented Programming, Cloud Computing, Containerization, Agile Development, Scrum, Accessibility (WCAG)

**Certifications:** Docker Foundations Professional Certificate | CIW Site Development Associate | IBM SkillsBuild AI Certifications (6x)

## PROJECTS:

### Personal Portfolio - [AWS-Hosted Static Website](#)

May 2026 - Present

- Developed a responsive portfolio website (20+ pages) using semantic HTML, CSS, and JavaScript with reusable components, dark mode support, and accessible navigation
- Designed and deployed cloud infrastructure using AWS S3, CloudFront, Cloudflare, GitHub Actions CI/CD, and IAM OIDC authentication

### Splatbot Game - [Coding Outreach](#)

Mar 2026 - Present

- Developed a browser-based competitive coding game to make programming accessible for JMU CS students
- Engineered a JavaScript game engine with SVG rendering and sandboxed in-browser Python bot execution using Pyodide and Web Workers to securely isolate user-submitted code
- Designed core gameplay systems, bot APIs, UI components, and persistent client-side settings
- Organized the inaugural Splatbot tournament fundraiser with 30+ participants, 50+ attendees, and 60+ matches
- Used Git and Jira for version control, sprint planning, feature tracking, and collaborative development workflows

### Jarvis AI Home Network - [24-Hour Hackathon Project](#)

Apr 2026

- Built a voice-driven AI assistant in under 24 hours, integrating Spotify, calendar, and smart-home systems
- Developed a Node.js/Express backend with REST APIs for scheduling, chat history, and device control
- Created a React frontend with an AI interface and dynamic system prompts
- Integrated Google Calendar API, Home Assistant webhooks, and LLM tool-calling workflows for automation

### **Social Network Platform for Spanish Speakers - Full Stack App**

**Jan 2026 - May 2026**

- Designed and implemented RESTful APIs for authentication, user management, and content interactions using Node.js and Express.js
- Developed relational database schemas and optimized SQL queries for user, post, and engagement data
- Implemented authentication, authorization, and secure credential management practices
- Deployed and maintained a production backend on Linux using Nginx and PM2 for a social platform serving 20+ university students

### **Terminal-Based Spreadsheet - Ruby Program**

**Jan 2025 - May 2025**

- Built a terminal-based spreadsheet in Ruby featuring a full lexer, parser, and formula evaluation engine supporting nested expressions, with interactive grid editing via Curses

### **Teaching Assistant Database - Web App**

**Aug 2024 - Dec 2024**

- Designed a MySQL database schema and built a Flask-based web interface for managing TA records
- Validated the system against 100+ records, demonstrating reliable CRUD operations and query performance across realistic data volumes

## **EXPERIENCE:**

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### **Independent Software Developer, JMU Pep Band**

**Sep 2025 - Present**

- Engineered a cross-platform attendance tracking system in Python used by 250+ members across 70+ events
- Integrated Google Sheets API to automate attendance validation, record categorization, and report generation workflows
- Packaged and distributed executables for macOS and Windows using PyInstaller
- Applied Agile/Scrum methodologies in Jira to manage sprint planning, task tracking, and project deliverables

### **VEX Robotics Mentor, Enginotic 6 Robotics**

**May 2019 - May 2022**

- Guided teams in mechanical design, C++ programming, problem-solving, competition strategy, and time management
- Mentored teams that qualified for State and World Championships

## **INVOLVEMENT:**

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### **CS Faculty Search Committee, JMU**

**Sep 2025 - May 2026**

#### *Student Representative*

- One of two undergraduates selected to participate in faculty candidate evaluation and interviews

### **Competitive Programming Club, JMU**

**Aug 2024 - May 2026**

#### *Social Chair: Aug 2025 - May 2026*

- Organized technical events and increased member engagement
- Solved algorithmic problems on platforms like Kattis

### **Hobbies/Interests - JMU Bands | Madison Motorsports | Competitive Gaming**